PERSONAL INFORMATION MANAGEMENT (PIM) SYSTEM

User Manual

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Group members

Name	Student ID
[Member 1 name]	[Student ID]
[Member 2 name]	[Student ID]
[Member 3 name]	[Student ID]
[Member 4 name]	[Student ID]

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1 Introduction

The PIM is a personal information manager that runs in a terminal. It keeps four kinds of personal information record (PIR) in one place:

Type	Fields	Required
Note	`text`, the body of the note	`text`
Task	`description`, `deadline`	`description`
Event	`description`, `start`, and any number of alarms	`description`, `start`
Contact	`name`, `address`, `mobile`	`name`

Every PIR has an **Id**, a number the system assigns when the record is created. The Id never changes, and it is never given to another record, even after a delete. You can save all your records in a file with the extension `.pim` and load them again later.

All times are in **Hong Kong Time (HKT)** unless you type another time-zone offset.

2 Starting the program

You need a Mac with Python 3.12 or newer. Open Terminal (or iTerm), go to the folder that contains `pim.py`, and run:

```
python3 pim.py
```

The program chooses its interface by itself:

- In an interactive terminal it opens the **full-screen interface**, which you work mostly with single keys, selectors, and a calendar.
- When its input or output is redirected (for example `python3 pim.py < commands.txt`), it uses the **line interface**: it prints the screen as text and reads one command per line. You can also ask for the line interface with `PIM_NO_CURSES=1 python3 pim.py`.

Both interfaces accept the same commands (Section 5). To leave the program, press `q` or type `quit`.

3 The screen

The full-screen interface has five areas, from top to bottom:

- **Title bar.** The name of the open file, or `untitled`, followed by `*` when there are unsaved changes, and the current time in HKT on the right.

- **Alarm band.** The first Event alarm that is OVERDUE or SOON, with `+N more` when there are others; `Alarms · none` when nothing is due (Section 9).
- **Current Result** (left). One row per record: the row number `#`, the Id, a type pin (`N`, `T`, `E`, or `C`), the Display Name, and the deadline or start. When a search is active, the pane title shows the criterion.
- **Details** (right). Every field of the selected record. The Id and type are in the pane title.
- **Composer** (bottom). Key hints when nothing is being asked, otherwise the current question: a selector, a calendar, a folder browser, or a text field. The status line, which reports the outcome of every command, sits just above it.

On a terminal narrower than 80 columns, the list is shown above the details. If the window is smaller than about 60 × 12 characters, the program asks you to enlarge it; `q` still quits. On a terminal with 256 colours the interface uses its own dark colour scheme; with fewer colours it uses reverse video for the title and the selected row.

The line interface prints the same information as plain text after every command:

```
PIM  untitled*
--------------------------------------------------------------------------------
ALARMS
  OVERDUE  Id 1  COMP3211 lecture  2026-09-13T18:30:00+08:00
  (dismiss)
--------------------------------------------------------------------------------
Current Result (all)
    #   Id  Type      Name                      Time
    1    1  event     COMP3211 lecture          2026-09-14T18:30:00+08:00
>   2    2  contact   Ada
--------------------------------------------------------------------------------
DETAIL
  Id: 2
  type: contact
  name: Ada
  address: (none)
  mobile: 12345678
--------------------------------------------------------------------------------
Created Contact Id 2
create  search  clear  modify  print  print all  delete  save  save as  load  dismiss  help  quit
>
```

The `>` in the list marks the selected row. The line below the last rule is the status line, followed by the list of commands and the prompt.

4 Using the full-screen interface

4.1 Keys

These keys act at once when no question is open:

Key	Action
↑ ↓ or `k` `j`	Select the previous or next row of the Current Result
`PgUp` `PgDn`, `Home` `End`	Move a page, or to the first or last row
`/`	Search: type one criterion line (Section 7). While the field is empty it shows an example. After a syntax error the typed line stays, so you can correct it
`c`	Create a record: choose the type, then answer the field questions
`m`	Modify the selected record
`p` / `P`	Print the selected record / print every record in the Current Result
`x` or `Delete`	Delete the selected record (asks Yes or No)
`d`	Dismiss the first alarm in the alarm band
`w`	Save (asks for a file when the collection has none yet)
`W`	Save as: choose a folder and a file name
`o`	Load (open) a `.pim` file
`:`	Type a full command, such as `id 4` or `save as work`
`?`	Show the key help; any key closes it
`Esc`	Cancel the current question or typed command; with nothing open, end the active search. It never discards unsaved changes
`q`	Quit

While a text field is open, type your answer and press Enter. On an optional field, Enter alone leaves it empty.

4.2 Selectors

When an answer is one of a few fixed values, the composer shows a **selector** instead of a text field, and you do not type the word. Move with ← → (or ↑ ↓, `Home`, `End`) and press Enter, or press the option's key, which confirms at once.

Question	Options and keys	Default
Type of a new record	Note `n` · Task `t` · Event `e` · Contact `c` (or `1`–`4`)	Note
Delete, overwrite, add an alarm, replace alarms	Yes `y` · No `n`	No
Alarm kind	Relative `r` · Absolute `a`	Relative
When a relative alarm rings	At start `0` · 15 minutes `1` · 1 hour `h` · 1 day `d` · Other… `o`	At start
Unit (after Other… and a number)	Minute `m` · Hour `h` · Day `d` · Week `w`	Day
Unsaved changes	Save `s` · Discard `d` · Cancel `c`	Cancel

`Esc` cancels a selector, except the unsaved-changes question: there, press `c` to cancel.

4.3 Entering dates and times

Event **start**, Task **deadline**, and the time of an **absolute alarm** open a calendar instead of a text field. The month grid on the left starts each week on Monday, and the time list on the right shows Hong Kong Time in 15-minute steps. The line under the grid shows the value that Enter will use, for example `Using  Tu 15 September 2026  18:30  HKT`.

Key	Action
← →	Grid: previous or next day. Time list: one hour earlier or later
↑ ↓	Grid: previous or next week. Time list: 15 minutes earlier or later
+ -	Time list: one minute later or earlier
Tab	Switch between the grid and the time list
[ ] (or PgUp PgDn)	Previous or next month
t	Jump to today, keeping the time
n	Leave an optional deadline empty, or clear it during modify
Enter	Use the highlighted date and time
Esc	Cancel the whole create or modify command

During modify, the calendar opens on the current value, so Enter keeps it. A Task without a deadline opens on the next quarter hour; press `n` to keep it without one.

In a **search** there is no calendar: type the time in the criterion, for example `start < 2026-11-20 20:00`.

4.4 Choosing a file

`o` (load) and `W` (save as) open a **folder browser**. It starts in the folder of the open file, or else in the folder from which you started the program, and lists the parent folder `../`, the subfolders, and the `.pim` files. Hidden files are not shown.

Key	Action
↑ ↓ or k j, PgUp PgDn, Home End or g G	Move the highlight
Enter, or → / l on a folder	Open the folder, or choose the highlighted `.pim` file (save as asks before replacing it)
Backspace, ←, or h	Go up one folder
~	Go to your home folder
/ or Tab	Type a path instead. Tab fills in the current folder and Ctrl-U clears the field. While the filled-in path is unchanged, Esc cancels at once; after you edit it, the first Esc clears the text and the second cancels
Esc	Cancel

To save under a new name, choose **+ New file in this folder…**, type the name, and press Enter; `.pim` is added for you. The PIM stays inside the terminal and never opens a Finder window.

4.5 Reading printed records

`p` and `P` open a PRINT panel over the screen. Scroll it with ↑ ↓ (or `k` `j`) and `PgUp` `PgDn`, and close it with `Esc`, Enter, or `q`. While the panel is open, `q` closes it rather than quitting.

5 Commands

You can type every command in the line interface, and after `:` in the full-screen interface.

Command	Action
`create` or `create note|task|event|contact`	Create a record. Without a type, the program asks for one. You are then asked for each field in turn.
`search <criterion>` or `search`	Show only the records that match the criterion (Section 7). Without a criterion, the program asks for one.
`clear`	End the search and show every record again.
`<n>` (a row number)	Select row `n` of the Current Result. Row numbers change with every search; they are not Ids.
`id <n>`	Select the record with Id `n`.
`modify`	Change the fields of the selected record (Section 6.2).
`print`	Print every field of the selected record.
`print all`	Print every record in the Current Result: all records when no search is active, otherwise the matches. Use `clear` first to print everything.
`delete`	Delete the selected record after you answer `y`. Its Id is not used again.
`save`	Save to the open file. With no open file, this works as `save as`.
`save as <path>` or `save as`	Save to a new file and keep working with it (Section 8).
`load <path>` or `load`	Replace the records in memory with those in a `.pim` file (Section 8).
`dismiss`	Hide the first alarm alert (Section 9).
`help`	Show the list of commands on the status line.
`quit` (also `q` or `exit`)	Leave the program. With unsaved changes, you are asked to save, discard, or cancel first.

An unknown command is reported as `unknown command: …` and changes nothing.

6 Creating and changing records

6.1 Creating

`create` asks for the fields of the chosen type one at a time, naming each field and saying whether it is optional. Leading and trailing spaces are removed. An empty answer, or one made only of spaces, leaves an optional field empty; on a required field it makes the create fail with a message such as `description is required`, and nothing is created.

Dates and times can be typed (in the line interface) as `YYYY-MM-DD HH:MM`, for example `2026-11-20 20:00`, or as an ISO 8601 instant such as `2026-11-20T12:00Z`. A time without an offset is Hong Kong Time, and seconds are ignored.

Alarms. After the start of an Event, you are asked `Add an alarm?`. Each alarm is one of two kinds:

- a **relative** alarm rings a whole number of minutes, hours, days, or weeks **before** the start, or exactly at the start. When the start changes, it moves with it;
- an **absolute** alarm rings at a fixed date and time, and does not move when the start changes.

After each alarm you are asked whether to add another. An Event may have no alarms at all.

6.2 Modifying

`modify` walks through the fields of the selected record and shows each current value in the question:

- press Enter alone to keep a value;
- on an optional field, type `none` (or only spaces) to clear it;
- type a new value to replace it.

For an Event you are asked `Replace alarms?`. Answer No to keep them. Answer Yes to build a new alarm list; answering No at the first `Add an alarm?` removes every alarm.

The Id and the type of a record never change. To turn a Note into a Task, delete the Note and create a Task. If a new value is invalid, the record stays exactly as it was. A modify that changes nothing reports `No changes` and does not count as an unsaved change.

7 Searching

A search criterion is one line made of tests, combined with `&&` (and), `||` (or), `!` (not), and parentheses:

```
type = note|task|event|contact
text|description|name|address|mobile contains "..."
contains "..."
deadline|start|alarm  <  |  >  |  =  <date and time>
```

- `contains` ignores upper and lower case and matches any part of the text; it does not do fuzzy matching. Without a field name, `contains` looks in every text field of the record.
- `<` means before, `>` means after, and `=` means at the same minute. Times are compared as instants, so `start = 2026-09-14T10:30Z` finds an Event that starts at 18:30 HKT.
- A comparison on a field that a record does not have, or has left empty, is false. A Task without a deadline never matches `deadline < …`.
- `alarm` matches an Event if **any** of its alarm times matches.
- `!` binds most tightly, then `&&`, then `||`. Use parentheses when in doubt.
- Keywords and field names may be written in any case. Inside quotation marks, write `\"` for a quotation mark and `\\` for a backslash.

Examples:

```
search type = event && description contains "COMP"
search contains "ada"
search type = task && !(deadline < 2026-12-01 00:00)
search alarm < 2026-09-14 12:00 || start > 2026-10-01 00:00
```

A successful search shows the matching records in Id order, selects the first one, and reports `3 match(es)`. A criterion that cannot be read is reported as `search syntax error: …`, and the list stays as it was. `clear`, or `Esc` in the full-screen interface, shows every record again.

8 Saving and loading

Paths. A path may be absolute (`/Users/you/Documents/work.pim`), start with `~` for your home folder (`~/Desktop/work`), or be relative to the folder from which you started the program. `.pim` is added when you leave it out, and missing folders are created. After a save or load, the status line shows the full path, for example `Saved /Users/you/Desktop/work.pim`.

Saving. `save as` replaces an existing file only after you confirm with Yes. The file is written safely: if the save fails, the previous file is left untouched and your changes stay unsaved.

Loading. `load` opens only files that end in `.pim`. If the file is not a valid PIM file, the program reports `not a PIM file: …` and keeps the records you had.

Unsaved changes. The `*` in the title bar means that some changes are not saved. Loading another file or quitting then asks you to **save**, **discard**, or **cancel**. Ctrl-C and Ctrl-D in an interactive terminal follow the same rule as quit, so you never lose changes without being asked. When the program reads its commands from a redirected file and the file ends while changes are unsaved, nobody can answer the question: the program ends without saving and prints `input ended: unsaved changes were not saved`.

9 Alarm alerts

While the program runs, it watches the alarms of every Event. An alarm is **OVERDUE** once its time has passed and **SOON** when it is due within the next 15 minutes. The alarm band shows the Event's Id, its description, and the time the alarm rings; it updates within half a second, even while you are not typing.

The line interface lists every due alarm. The full-screen interface shows the first one and the number of others. `d` or `dismiss` hides the first alarm until you quit the program. An alarm you dismissed appears again after you load a file or change that Event's alarms. Dismissals are not saved in the file.

10 Reading the output

Every command ends with one line on the **status line**: what happened (`Created Event Id 1`, `3 match(es)`, `Saved /Users/you/Desktop/work.pim`) or what went wrong (Section 11). In the full-screen interface, a success is shown in green and an error in red.

The **Display Name** in the list is the first line of a Note, the description of a Task or Event, and the name of a Contact. The time column shows a Task's deadline or an Event's start.

`print` and `print all` show every field. Times are ISO 8601 with their offset, where `+08:00` is Hong Kong Time, and an optional field that is not set shows `(none)`. Each Event alarm shows its kind and the **effective** time at which it rings:

```
Id: 1
type: event
description: COMP3211 lecture
start: 2026-09-14T18:30:00+08:00
alarm[0]: relative 1 day before start; effective 2026-09-13T18:30:00+08:00
alarm[1]: absolute; effective 2026-09-13T09:00:00+08:00

Id: 2
type: contact
name: Ada
address: (none)
mobile: 12345678
```

11 Error messages

An invalid command never changes your records, never shows a program error trace, and never closes the program. The status line names the problem:

Status line	Meaning and what to do
`text is required` (also `description`, `start`, `name`, `alarm time`)	A required field was left empty. Run the command again and fill it in.

Status line	Meaning and what to do
`invalid datetime: …`	The value is not a date and time in the form `YYYY-MM-DD HH:MM` or ISO 8601.
`datetime out of range`	The time falls outside the years 1–9999 in Hong Kong Time or UTC, for example `0001-01-01 00:10`.
`alarm time is out of range`	A relative alarm would ring before the year 1, for example after a huge number of weeks.
`relative alarm amount must be an integer`	Enter a whole number such as `15`, not `1.5`.
`relative alarm cannot be after start`	A relative alarm needs a number of 0 or more; use an absolute alarm for a time after the start.
`relative alarm unit must be minute, hour, day, or week`	The unit was not one of these four.
`unknown PIR type: …`	`create` accepts only `note`, `task`, `event`, or `contact`.
`no PIR selected`	`modify`, `print`, or `delete` needs a selected record. Select a row first.
`no PIR with Id …`	No record has this Id.
`no row … in Current Result`	The row number is outside the list on screen.
`search syntax error: …`	The criterion could not be read; the list is unchanged. A bad time in a criterion is reported as `search syntax error: invalid datetime: ….`
`path is required`	`save as` or `load` was given an empty path.
`file name is required`	The path has no file name, for example only `.pim` or a folder ending in `/`.
`path must have a .pim extension`	`load` opens only `.pim` files.
`not a PIM file: …`	The file is missing, unreadable, or not a valid PIM file. Your records in memory are unchanged.
`cannot save …: …`	The operating system refused the write, for example because permission was denied. Your changes stay unsaved.
`unknown command: …`	The command does not exist; type `help` for the list.
`enter y or n`, `enter relative or absolute`, `enter save, discard, or cancel`	In the line interface, the answer did not match the question. The question is asked again.
`command failed`	An unexpected internal error. The command did nothing, and the program keeps running.

12 Example sessions

Full-screen interface. Create a Note, find the lecture Event, and change it:

```
c                     open the type selector
n                     choose Note
Shopping: Milk        type the text and press Enter
/                     open the search field
type = event && description contains ”COMP”     then Enter
m                     modify the selected Event
```

Line interface. The same kind of work as typed commands, one per line (in the full-screen interface, type `:` before each command):

```
create note
Shopping: Milk
create event
COMP3211 lecture
2026-09-14 18:30
y
relative
1 day
n
search type = event && description contains ”COMP”
print
clear
print all
save as demo
quit
```

The first answer after `create event` is the description, then the start, then the alarm questions: `y` adds an alarm, `relative` and `1 day` describe it, and `n` stops adding alarms.